

QuizSmart - Program Architecture

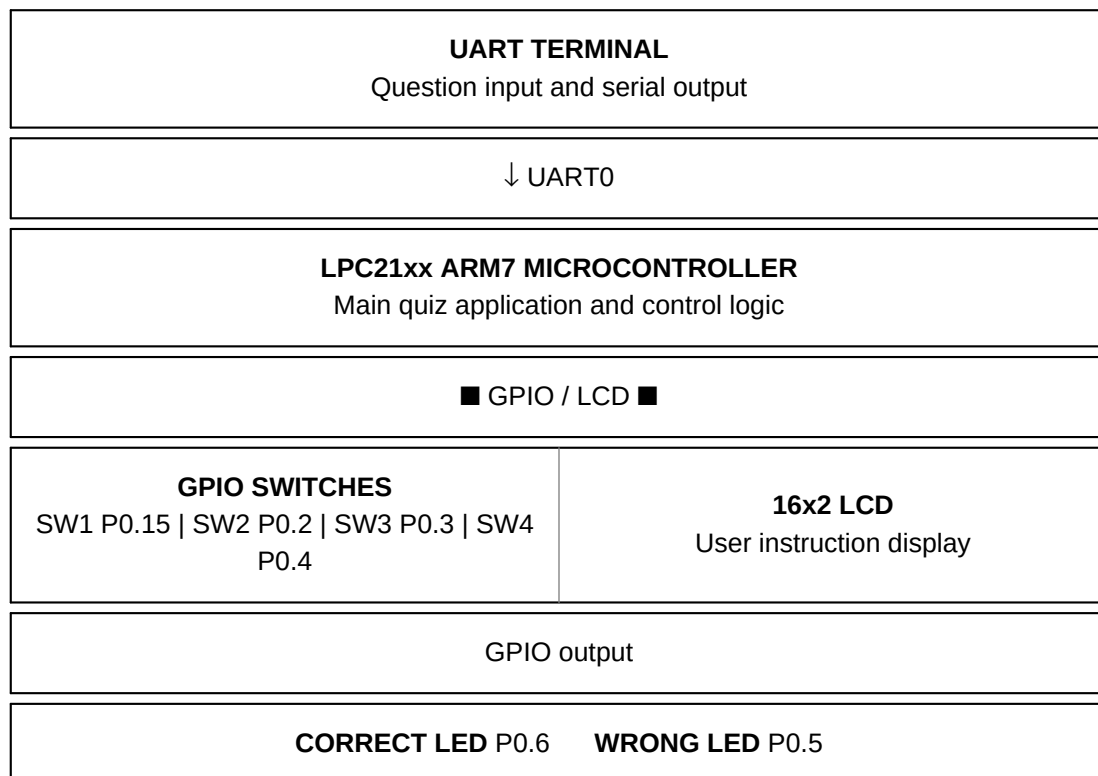
ARM7 LPC21xx Embedded Interactive Quiz System

System Architecture

QuizSmart follows a modular embedded-system architecture in which the LPC21xx ARM7 microcontroller acts as the central controller. UART0 provides the serial user interface, GPIO handles push-button inputs and LED outputs, and the LCD provides local user instructions.

Module	Interface	Responsibility
UART Terminal	UART0	Question menu, question/options display, question-number input, result display
Four Push Buttons	GPIO inputs	User answer selection: SW1=A, SW2=B, SW3=C, SW4=D
16x2 LCD	LCD Driver	Displays selection instruction to the user
Correct/Wrong LEDs	GPIO outputs	Hardware result indication interface
LPC21xx ARM7	Main Application	Peripheral initialization, quiz control, option detection and answer verification

Block Architecture



Pin Configuration

Signal	LPC21xx Pin	Purpose
SW1	P0.15	Option 1 / A
SW2	P0.2	Option 2 / B
SW3	P0.3	Option 3 / C
SW4	P0.4	Option 4 / D
Wrong LED	P0.5	Incorrect answer indication
Correct LED	P0.6	Correct answer indication

Modular Software Structure

Application Layer: main.c contains the quiz menu, question handling, answer verification and user interaction logic.

Driver Layer: UART, LCD, GPIO and delay routines can be maintained as separate reusable modules.

Hardware Layer: LPC21xx ARM7, UART interface, LCD, push buttons and LEDs.

Documentation: Individual driver and delay documentation can be maintained separately in the repository documentation directory.